

Drakens Energy 360

An end-to-end synthetic data engineering platform for a downstream energy business

Report generated 2026-09-07 · Warehouse: drakens360_test_full.duckdb

Independent synthetic portfolio project. Drakens Energy is a fictional company invented for this project. It does not exist. Every site, customer, employee, supplier, asset, coordinate, price, volume, margin and incident in this report is randomly generated. Site coordinates are South African town centroids plus random jitter and do not represent any real service station.

Summary

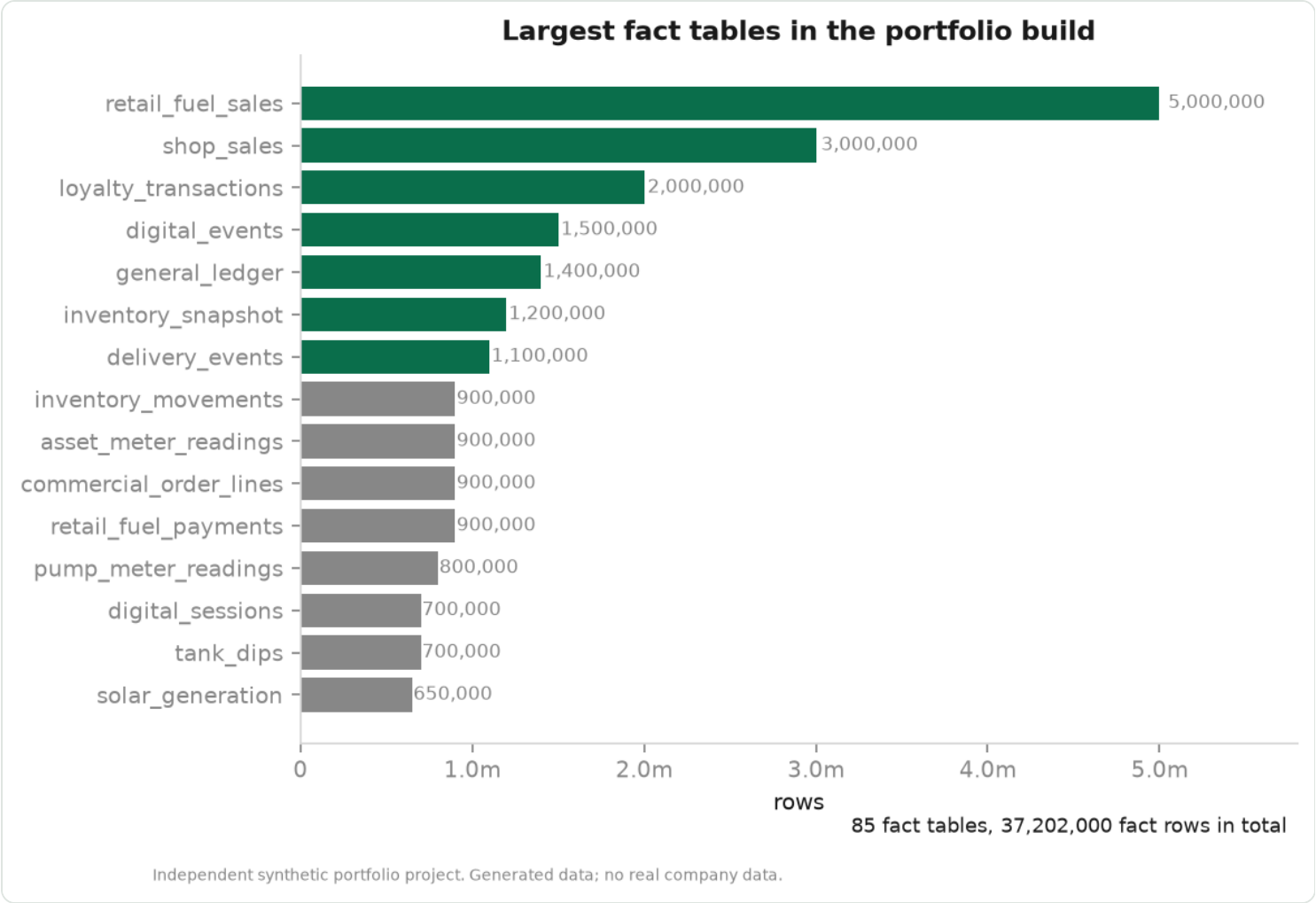
FACT TABLES	FACT ROWS	DEFECTS INJECTED	PROVINCES
85	37,202,000	62,924,526	9

The platform covers retail forecourt, commercial B2B, supply and distribution, LPG, lubricants, aviation, marine, loyalty, digital, EV and solar, asset maintenance, HSSEQ and finance — modelled on South Africa with all nine provinces and roughly 1,050 synthetic service stations. It is built to demonstrate the parts of data engineering that are hard, not the parts that demo well: a deliberately dirty landing zone, a cleansing layer whose output is measured rather than asserted, and models reported against baselines they have to beat.

Build

Table	Rows
fact_retail_fuel_sales	5,000,000
fact_shop_sales	3,000,000
fact_loyalty_transactions	2,000,000
fact_digital_events	1,500,000
fact_general_ledger	1,400,000
fact_inventory_snapshot	1,200,000
fact_delivery_events	1,100,000
fact_inventory_movements	900,000
fact_asset_meter_readings	900,000
fact_commercial_order_lines	900,000
fact_retail_fuel_payments	900,000
fact_pump_meter_readings	800,000
fact_digital_sessions	700,000
fact_tank_dips	700,000
fact_solar_generation	650,000
fact_customer_invoices	620,000
fact_site_energy_consumption	600,000
fact_customer_receipts	560,000
fact_commercial_orders	550,000
fact_mobile_payments	520,000
fact_site_daily_operations	500,000
fact_loyalty_points_balance	480,000

Table	Rows
fact_finance_revenue_cost	420,000
fact_forecourt_shift	420,000
fact_lpg_sales	400,000

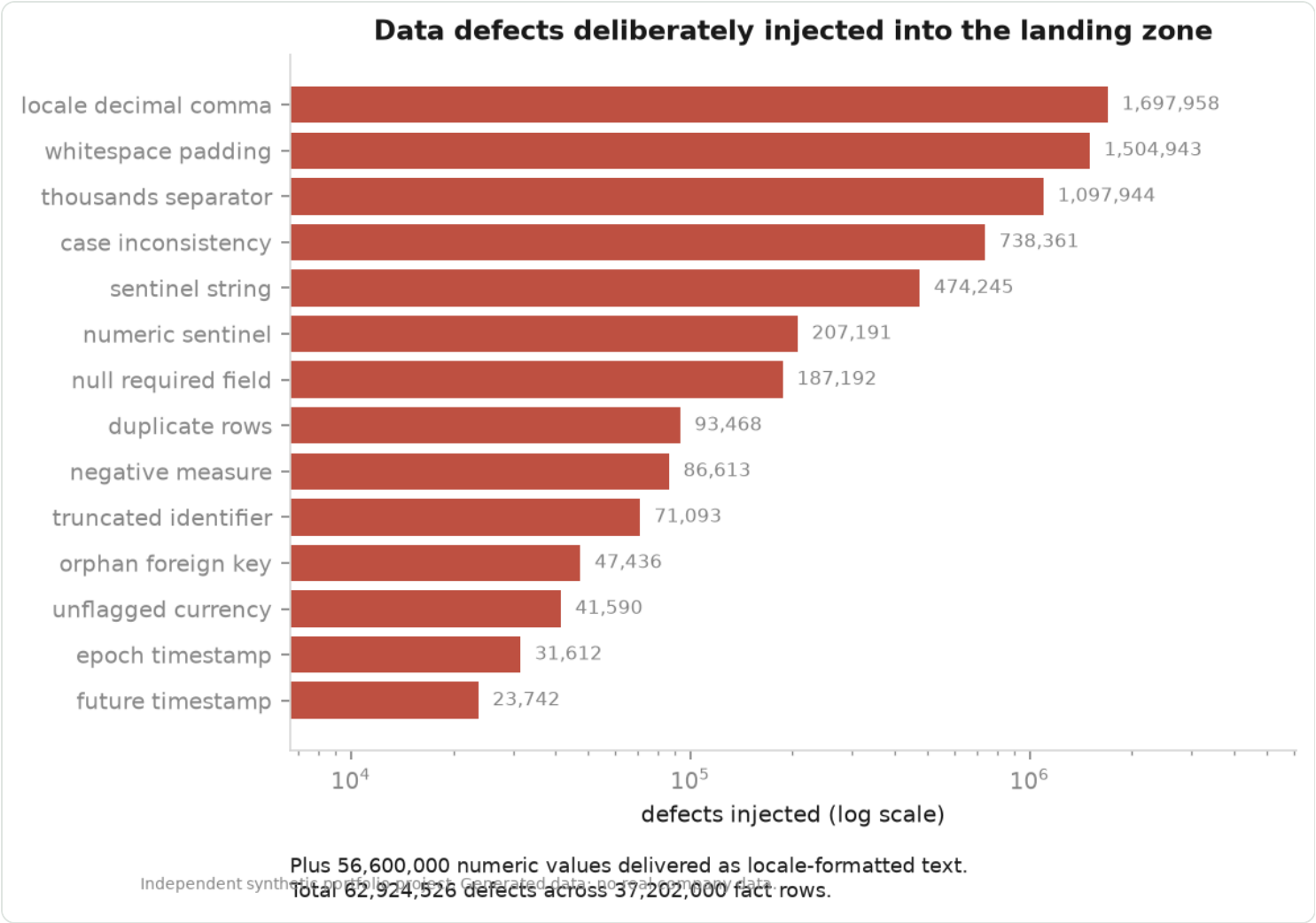


Largest fact tables in the portfolio build.

Data quality

The landing zone is dirty on purpose. A portfolio project built on perfect data demonstrates nothing, because the difficult part of the job is deciding what to do with the row where litres reads "1 234,5", the site code has a trailing space, and the same transaction arrived twice because the point-of-sale replayed its batch. Every defect is counted as it is injected, which turns cleansing from an assertion into a measurement.

Defect type	Defects
numeric delivered as text	56,600,000
locale decimal comma	1,697,958
whitespace padding	1,504,943
thousands separator	1,097,944
case inconsistency	738,361
sentinel string	474,245
numeric sentinel	207,191
null required field	187,192
duplicate rows	93,468
negative measure	86,613
truncated identifier	71,093
orphan foreign key	47,436
unflagged currency	41,590
epoch timestamp	31,612
future timestamp	23,742
derived value mismatch	12,256
unit of measure error	7,233
encoding mojibake	1,649



Defects deliberately injected into the landing zone.

Cleansing outcome

Table	Raw rows	Cleansed	Quarantined	Duplicates removed
fact_retail_fuel_sales	200,000	194,933	4,572	495
fact_shop_sales	120,000	116,702	2,992	306
fact_general_ledger	84,000	83,256	540	204
fact_loyalty_transactions	80,000	78,944	857	199
fact_delivery_events	66,000	65,557	283	160
fact_digital_events	60,000	59,593	236	171
fact_retail_fuel_payments	54,000	53,382	489	129
fact_commercial_order_lines	54,000	53,145	728	127
fact_asset_meter_readings	54,000	53,453	397	150
fact_inventory_movements	54,000	53,201	656	143
fact_inventory_snapshot	50,000	49,300	583	117
fact_pump_meter_readings	48,000	47,383	502	115
fact_tank_dips	42,000	41,351	543	106
fact_digital_sessions	42,000	41,745	162	93
fact_customer_invoices	37,200	36,733	374	93
fact_site_energy_consumption	36,000	35,623	305	72
fact_customer_receipts	33,600	33,205	301	94
fact_mobile_payments	31,200	30,842	284	74
fact_site_daily_operations	30,000	29,556	365	79
dim_loyalty_member	30,000	30,000	0	0
fact_loyalty_points_balance	28,800	28,491	232	77
fact_solar_generation	26,000	25,704	228	68
fact_finance_revenue_cost	25,200	22,476	2,662	62
fact_forecourt_shift	25,200	24,915	217	68

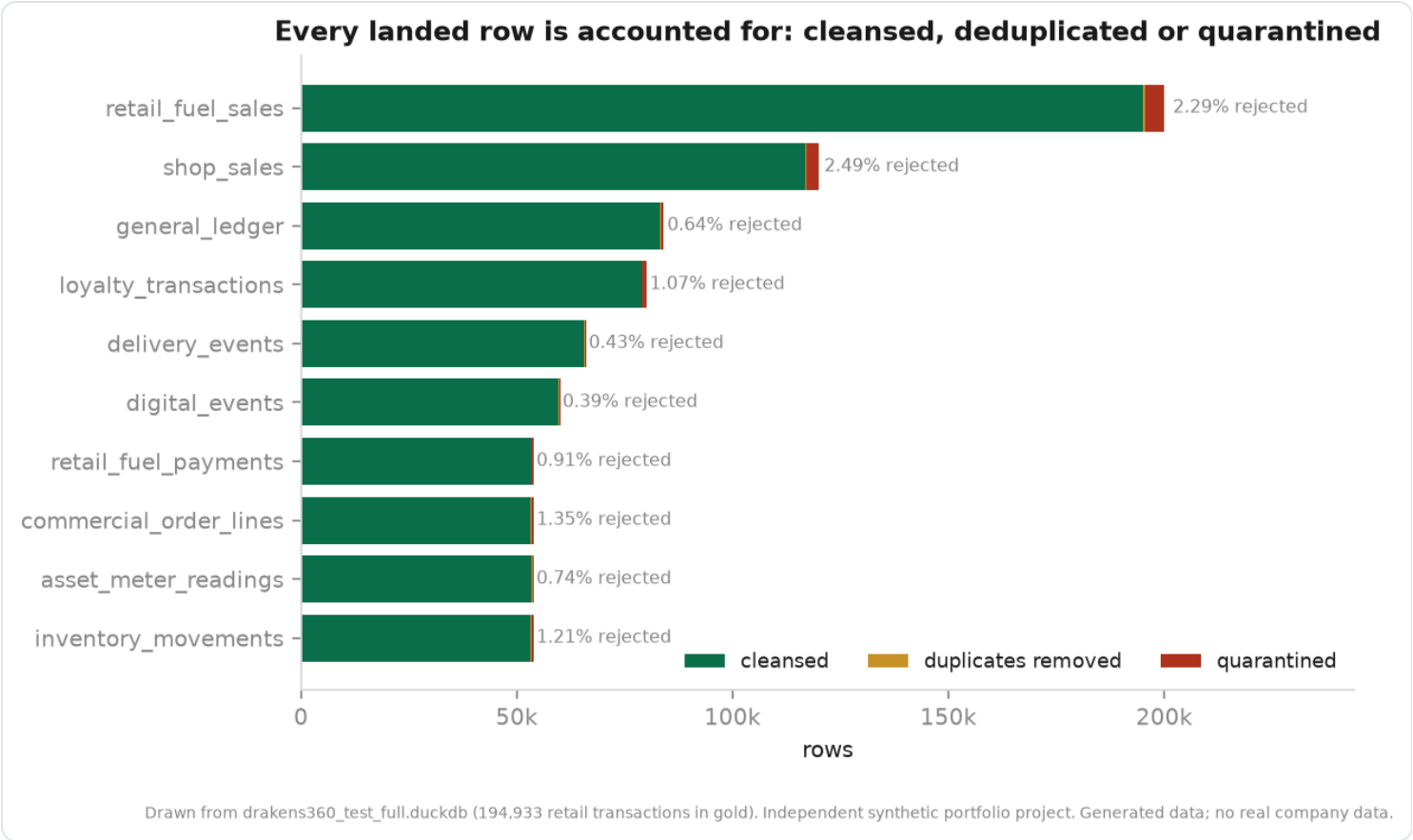
Table	Raw rows	Cleansed	Quarantined	Duplicates removed
dim_digital_user	25,000	25,000	0	0
fact_commercial_orders	25,000	24,417	528	55
fact_ev_charger_status	24,000	23,737	199	64
fact_grid_import_export	22,800	22,586	152	62
fact_commercial_deliveries	22,800	22,447	293	60
fact_accounts_receivable	22,800	22,508	232	60
fact_daily_executive_kpi	22,800	22,385	357	58
fact_vehicle_trips	20,400	20,247	100	53
fact_opex	19,200	18,995	164	41
fact_customer_credit_exposure	18,000	17,735	234	31
fact_accounts_payable	18,000	17,848	102	50
fact_forecast	18,000	17,739	222	39
fact_lpg_cylinder_movements	18,000	17,814	143	43
fact_lpg_sales	16,000	15,627	331	42
fact_price_changes	15,600	15,353	204	43
fact_budget	15,600	15,479	87	34
fact_replenishment_orders	15,600	15,362	192	46
fact_campaign_responses	15,600	15,470	92	38
fact_training_compliance	15,600	15,442	123	35
fact_deliveries	15,000	14,796	175	29
fact_budget_vs_actual	14,400	14,262	102	36
fact_lubricant_sales	14,000	13,687	281	32
fact_terminal_issues	12,600	12,463	95	42
fact_asset_inspections	12,000	11,879	93	28

Table	Raw rows	Cleansed	Quarantined	Duplicates removed
fact_spare_parts_usage	11,400	11,292	83	25
fact_procurement_orders	11,400	11,268	110	22
fact_safety_observations	10,800	10,715	55	30
fact_fleet_costs	10,800	10,704	59	37
fact_terminal_throughput	10,800	10,686	93	21
fact_procurement_receipts	10,000	9,858	109	33
fact_energy_costs	9,600	9,470	111	19
fact_lubricant_orders	9,000	8,882	94	24
fact_route_performance	9,000	8,926	56	18
fact_permit_to_work	9,000	8,910	73	17
bridge_customer_contract	8,947	6,000	0	2,947
fact_terminal_receipts	8,400	8,314	64	22
fact_aircraft_uplifts	7,800	7,729	49	22
fact_lpg_inventory_snapshot	7,200	7,126	60	14
fact_contract_pricing	7,200	7,123	62	15
fact_cashflow	7,200	7,125	61	14
fact_asset_downtime	7,200	7,117	69	14
fact_inventory_adjustments	7,200	7,110	74	16
fact_lubricant_inventory_snapshot	6,600	6,513	75	12
dim_customer	6,000	5,852	148	0
bridge_customer_hierarchy	6,000	300	0	5,700
fact_ev_charging_sessions	6,000	5,886	102	12
fact_working_capital	5,400	5,337	44	19
fact_vehicle_safety_events	5,400	5,352	34	14

Table	Raw rows	Cleansed	Quarantined	Duplicates removed
fact_contract_volume_commitments	5,400	5,337	50	13
fact_aviation_sales	5,000	4,897	88	15
dim_asset	5,000	5,000	0	0
bridge_asset_site	5,000	5,000	0	0
bridge_employee_site	4,256	3,000	0	1,256
fact_customer_service_cases	4,200	4,154	38	8
fact_asset_failures	4,200	4,142	46	12
fact_lpg_bulk_deliveries	4,200	4,155	33	12
fact_marine_sales	4,000	3,921	71	8
dim_pump	4,000	4,000	0	0
fact_maintenance_work_orders	4,000	3,958	38	4
fact_stock_losses	3,600	3,550	35	15
fact_shop_returns	3,600	3,548	41	11
fact_bunker_deliveries	3,600	3,571	18	11
bridge_site_product	3,536	600	0	2,936
dim_contract	3,500	3,500	0	0
bridge_customer_site	3,172	1,500	0	1,672
dim_employee	3,000	3,000	0	0
fact_near_misses	2,760	2,729	24	7
dim_tank	2,500	2,500	0	0
fact_capex	2,400	2,369	26	5
fact_promotions	2,400	397	223	1,780
fact_aviation_inventory_snapshot	2,400	2,365	27	8
bridge_route_site	1,869	600	0	1,269

Table	Raw rows	Cleansed	Quarantined	Duplicates removed
fact_marine_inventory_snapshot	1,800	1,776	16	8
fact_environmental_events	1,320	1,296	18	6
dim_equipment	1,250	1,250	0	0
bridge_supplier_product	1,206	400	0	806
dim_driver	1,100	1,100	0	0
bridge_site_promotion	1,018	600	0	418
dim_vehicle	900	900	0	0
fact_hsseq_incidents	800	795	4	1
bridge_site_hierarchy	600	586	14	0
dim_route	600	600	0	0
dim_site	600	586	14	0
fact_import_cargoes	540	538	2	0
dim_ev_charger	400	400	0	0
dim_supplier	400	385	15	0
dim_solar_asset	300	300	0	0
dim_city	88	88	0	0
dim_terminal	60	60	0	0
dim_promotion	44	44	0	0
dim_depot	40	40	0	0
dim_campaign	38	38	0	0
dim_product	31	31	0	0
bridge_product_hierarchy	31	31	0	0
dim_price_zone	30	30	0	0
dim_country	29	28	1	0

Table	Raw rows	Cleansed	Quarantined	Duplicates removed
dim_currency	25	25	0	0
dim_carrier	22	22	0	0
dim_warehouse	20	20	0	0
dim_airport	12	12	0	0
dim_legal_entity	12	12	0	0
dim_province	9	9	0	0
dim_port	8	8	0	0
dim_lpg_cylinder_type	4	4	0	0



Every landed row is accounted for as cleansed, deduplicated or quarantined.

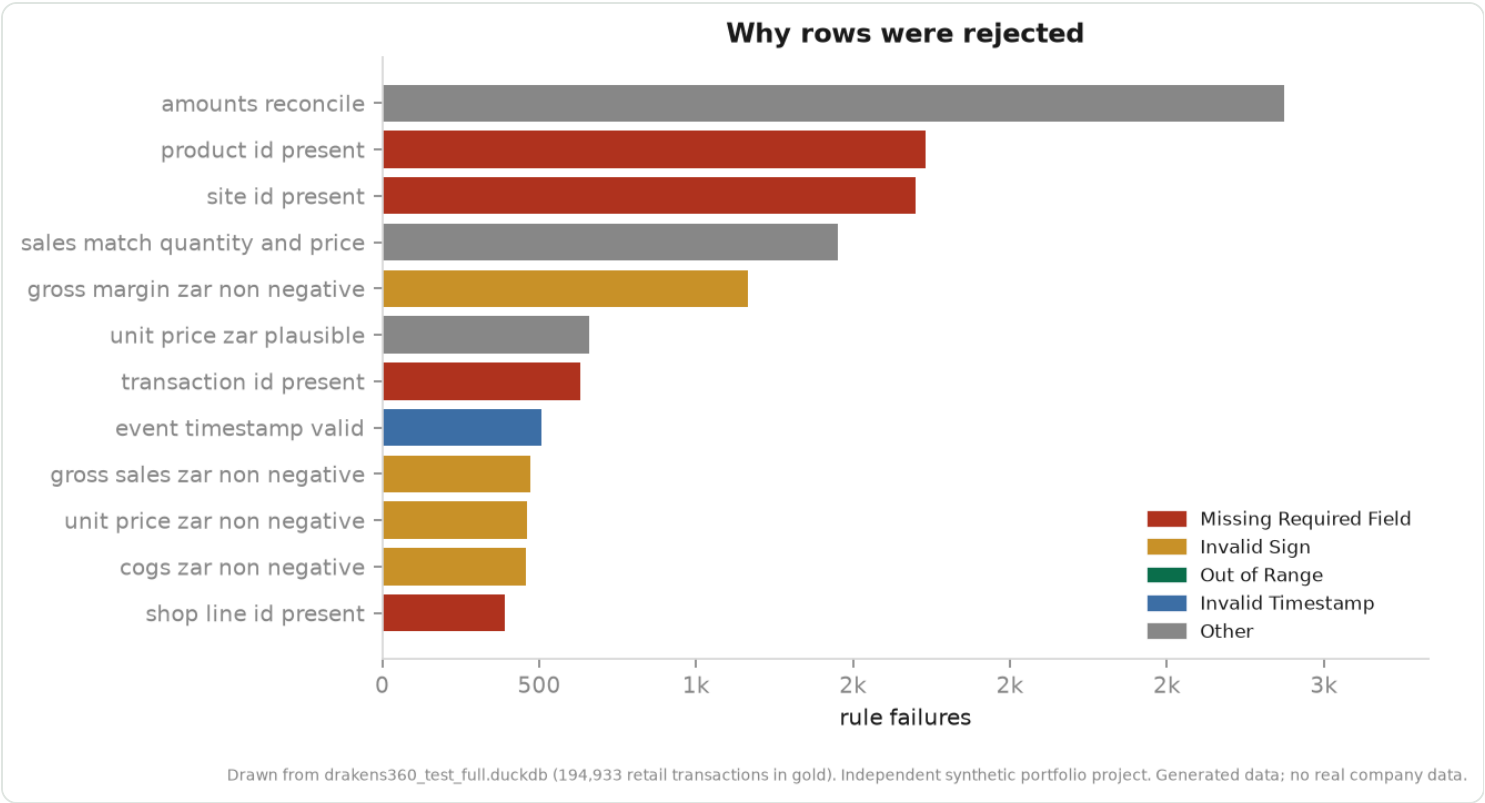
Why rows were rejected

Rule that rejected the row	Category	Likely owner	Rows
amounts_reconcile	Other	Data Platform	2,873
product_id_present	Missing Required Field	Source System	1,731
site_id_present	Missing Required Field	Source System	1,699
sales_match_quantity_and_price	Other	Data Platform	1,452
gross_margin_zar_non_negative	Invalid Sign	Business Process	1,166
unit_price_zar_plausible	Other	Data Platform	660.00
transaction_id_present	Missing Required Field	Source System	630.00
event_timestamp_valid	Invalid Timestamp	Source System	506.00
gross_sales_zar_non_negative	Invalid Sign	Business Process	473.00
unit_price_zar_non_negative	Invalid Sign	Business Process	460.00
cogs_zar_non_negative	Invalid Sign	Business Process	458.00
shop_line_id_present	Missing Required Field	Source System	391.00
journal_line_id_present	Missing Required Field	Source System	311.00
loyalty_transaction_id_present	Missing Required Field	Source System	284.00
litres_non_negative	Invalid Sign	Business Process	253.00
digital_event_id_present	Missing Required Field	Source System	197.00
reading_id_present	Missing Required Field	Source System	176.00
litres_within_range	Out of Range	Data Entry	161.00
snapshot_id_present	Missing Required Field	Source System	160.00
fill_pct_in_range	Other	Data Platform	143.00
tank_dip_id_present	Missing Required Field	Source System	125.00
points_value_zar_non_negative	Invalid Sign	Business Process	117.00
debit_amount_zar_non_negative	Invalid Sign	Business Process	93.00
order_id_present	Missing Required Field	Source System	91.00

Rule that rejected the row	Category	Likely owner	Rows
solar_reading_id_present	Missing Required Field	Source System	91.00
qualifying_spend_zar_non_negative	Invalid Sign	Business Process	90.00
credit_amount_zar_non_negative	Invalid Sign	Business Process	88.00
volume_litres_non_negative	Invalid Sign	Business Process	88.00
customer_id_present	Missing Required Field	Source System	88.00
capacity_litres_non_negative	Invalid Sign	Business Process	83.00
stock_on_hand_litres_non_negative	Invalid Sign	Business Process	69.00
volume_litres_within_range	Out of Range	Data Entry	59.00
delivery_id_present	Missing Required Field	Source System	56.00
revenue_zar_non_negative	Invalid Sign	Business Process	43.00
energy_kwh_non_negative	Invalid Sign	Business Process	39.00
ev_session_id_present	Missing Required Field	Source System	33.00
energy_kwh_within_range	Out of Range	Data Entry	32.00
fill_rate_pct_in_range	Other	Data Platform	28.00
list_price_zar_non_negative	Invalid Sign	Business Process	27.00
delivered_litres_non_negative	Invalid Sign	Business Process	19.00
delivered_litres_within_range	Out of Range	Data Entry	13.00
grid_cost_zar_non_negative	Invalid Sign	Business Process	8.00
work_order_id_present	Missing Required Field	Source System	8.00
total_cost_zar_non_negative	Invalid Sign	Business Process	7.00
labour_cost_zar_non_negative	Invalid Sign	Business Process	7.00
rated_power_kw_non_negative	Invalid Sign	Business Process	7.00
average_power_kw_non_negative	Invalid Sign	Business Process	3.00
parts_cost_zar_non_negative	Invalid Sign	Business Process	3.00

Rule that rejected the row	Category	Likely owner	Rows
incident_id_present	Missing Required Field	Source System	2.00

estimated_cost_zar_non_negative	Invalid Sign	Business Process	1.00
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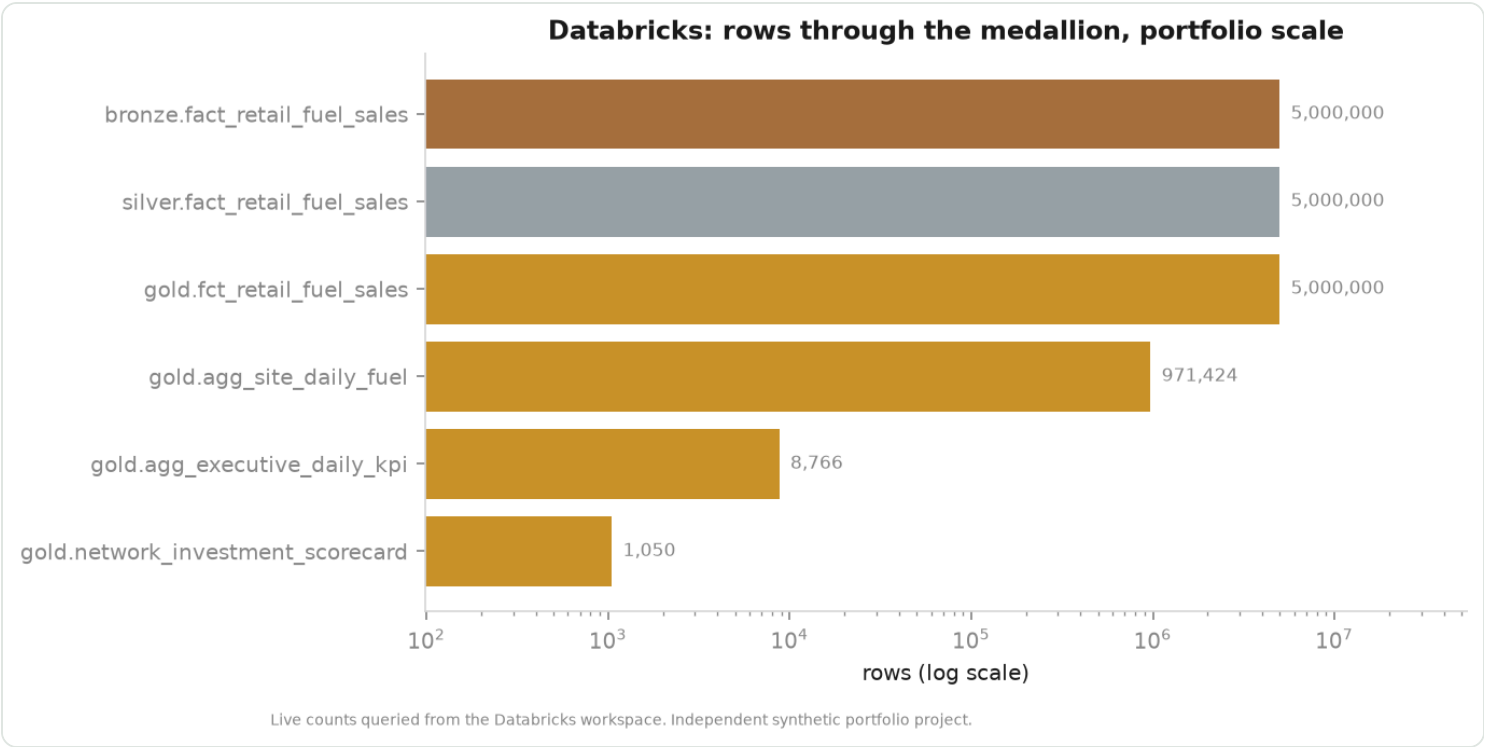


Rejection reasons. The operational response differs by rule.

Warehouse

Control	What it checks	Owner	Variance %	Result
DELIVERIES_HAVE_KNOWN_DESTINATION	Every delivery resolves to a site in the conformed dimension	Supply	0.00	PASS
GL_DEBITS_EQUAL_CREDITS	Posted journal debits equal credits across the ledger	Finance	0.00	PASS
ORDERS_HAVE_KNOWN_CUSTOMER	Every commercial order resolves to a customer in the dimension	Commercial	0.00	PASS
RETAIL_MARGIN_RECONCILES	Retail fuel gross margin equals revenue less cost of sales	Finance	0.00	PASS
RETAIL_VOLUME_TIES_TO_DAILY_AGG	Transaction-level litres tie to the site-day aggregate	Supply	0.00	PASS

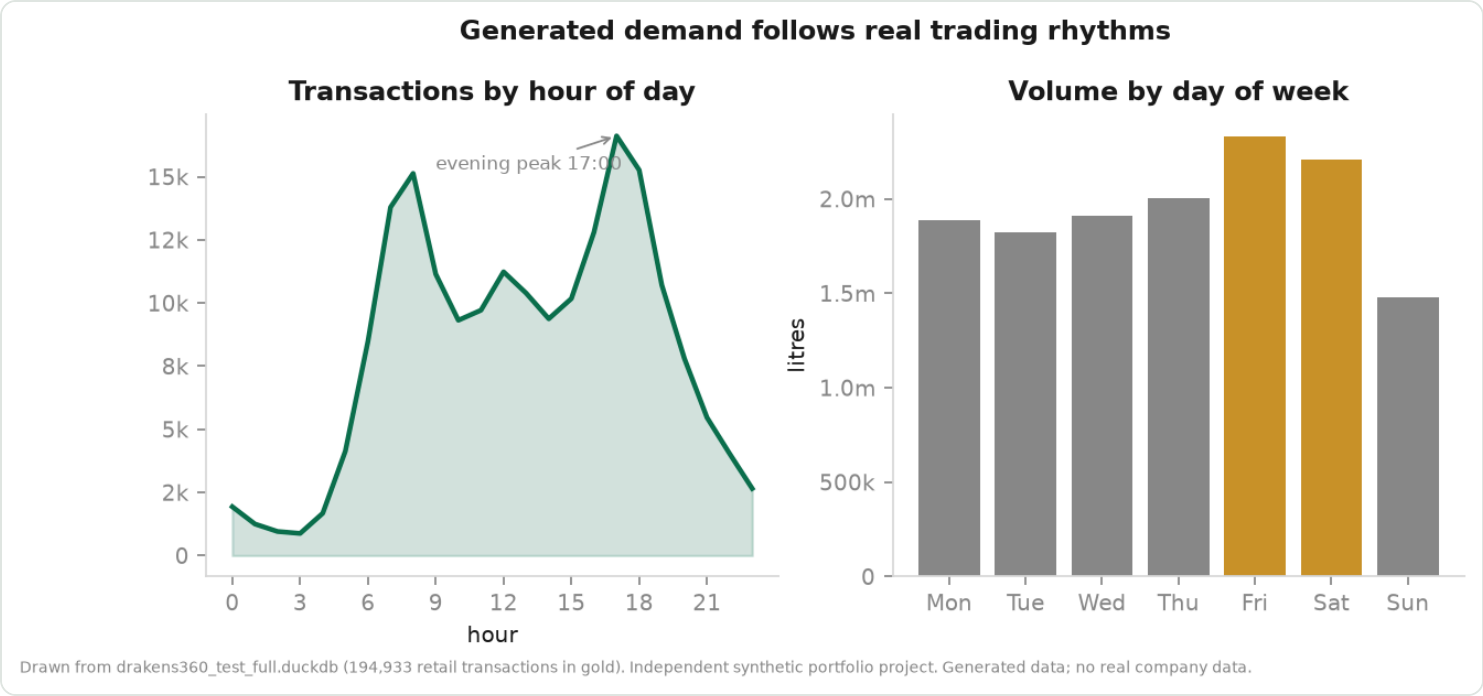
A reconciliation control evaluates a published number as data rather than only as a test, so a control that fails names the figure that is wrong and by how much. Controls that fail are shown as failures; a report that quietly omits them would defeat the purpose of having them.



Live row counts through bronze, silver and gold on Databricks at portfolio scale.

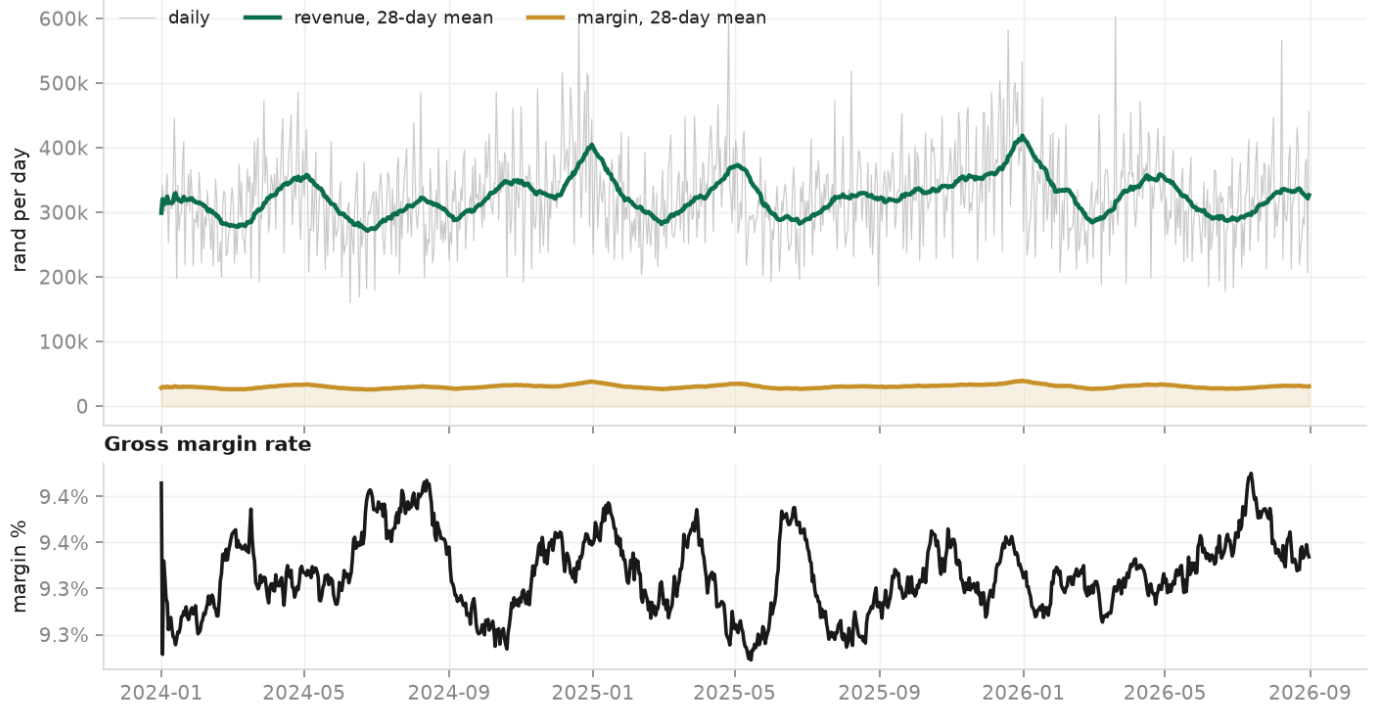
Demand and trading rhythm

A forecasting model can only learn a pattern the data actually contains. These three figures are the evidence that the generator produces one: an hourly commute shape, a weekly rhythm, and a southern-hemisphere seasonal cycle that survives all the way through to the warehouse.



Commute peaks at 08:00 and 17:00, and a Friday-heavy week.

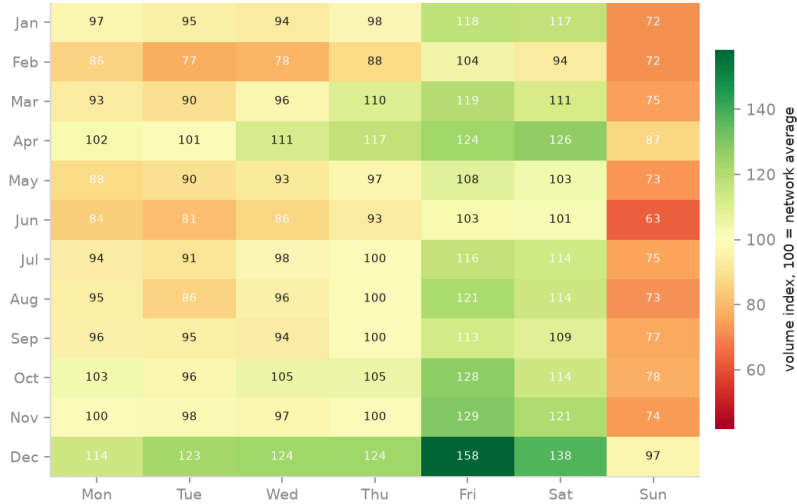
Network revenue and margin



Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Revenue and margin across the modelled window, with the margin rate on its own axis. A rate and a total share an axis only in charts that say nothing about either.

Volume by month and day of week

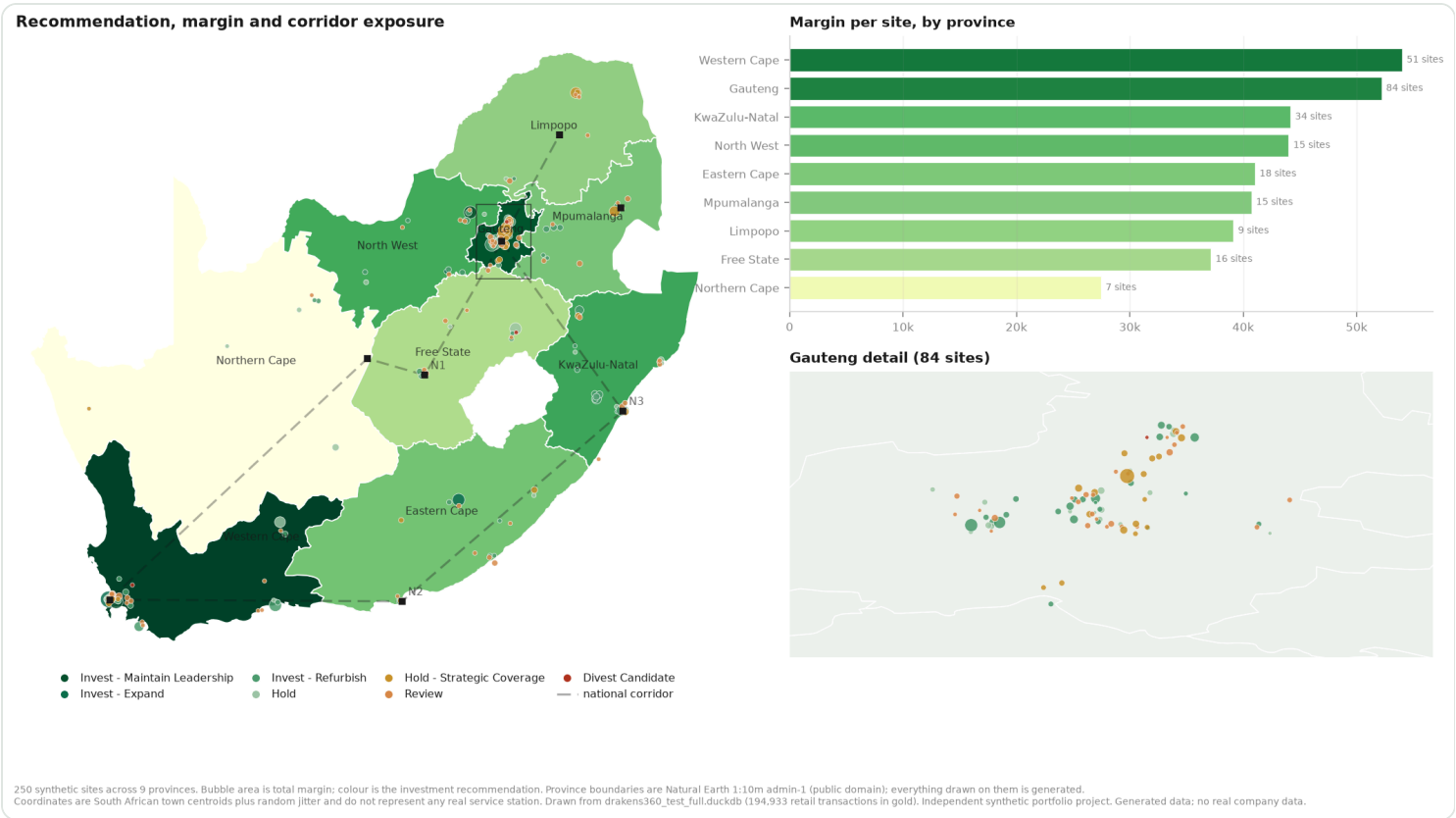


Mean litres per trading day, indexed to 100. Per day rather than per month because the modelled window ends in August: summed, September to December would read a third low and look like seasonality.

Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Month against day of week, indexed to the network average. December and January are the high-summer holiday peak; June and July the winter trough.

Network and investment

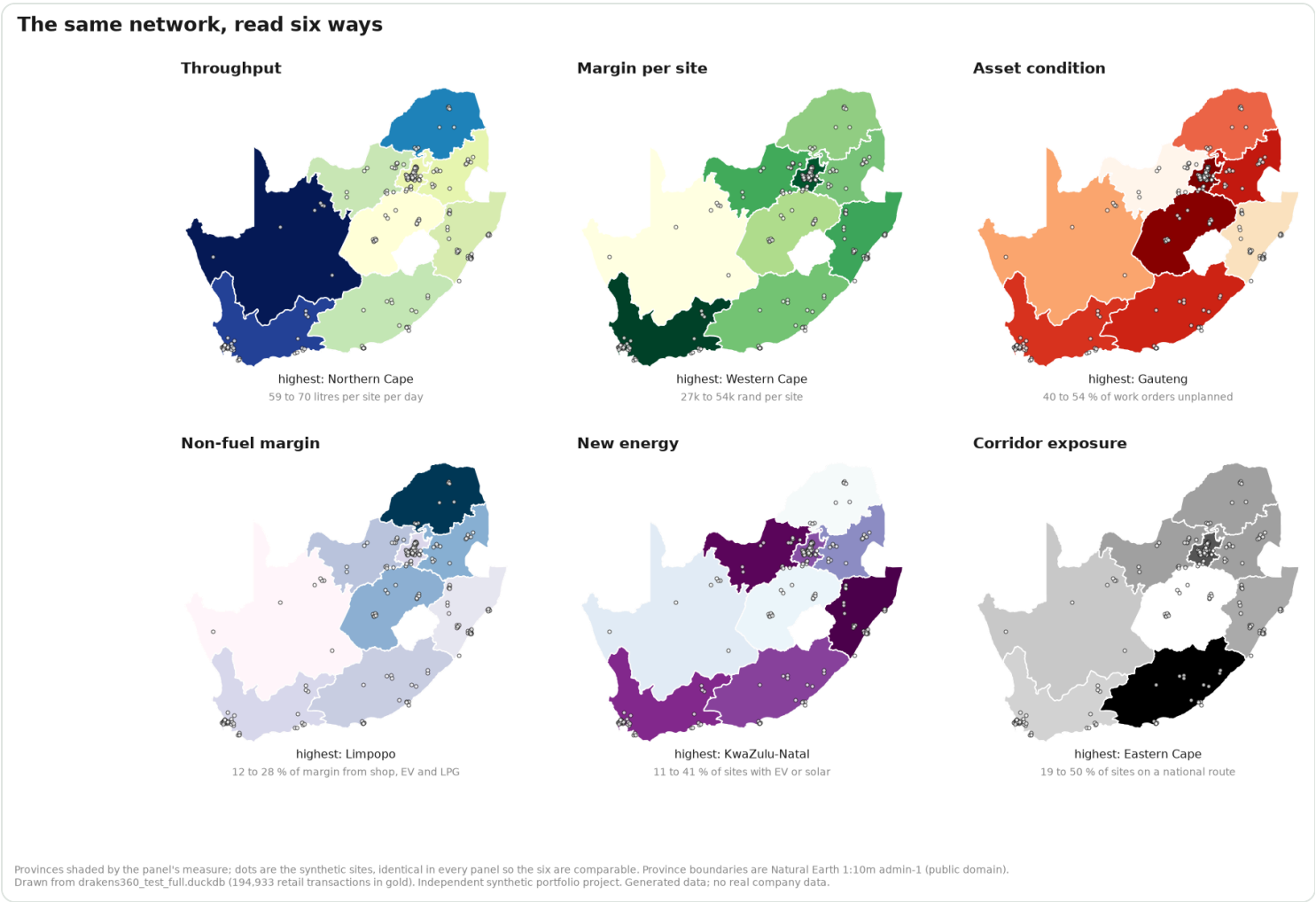


Provinces shaded by margin per site; bubble area is total margin and colour is the investment recommendation. The N1, N2 and N3 are drawn because corridor sites are protected from divestment regardless of score. Province boundaries are Natural Earth 1:10m admin-1, public domain; everything drawn on them is generated.

Measure	Value
Budget	R1.50bn
Funded projects	248
Capital committed	R1.330bn
Expected annual uplift	R332.9m
Portfolio ROI	25.0% per year
Blended payback	4.0 years
Provinces covered	9 of 9

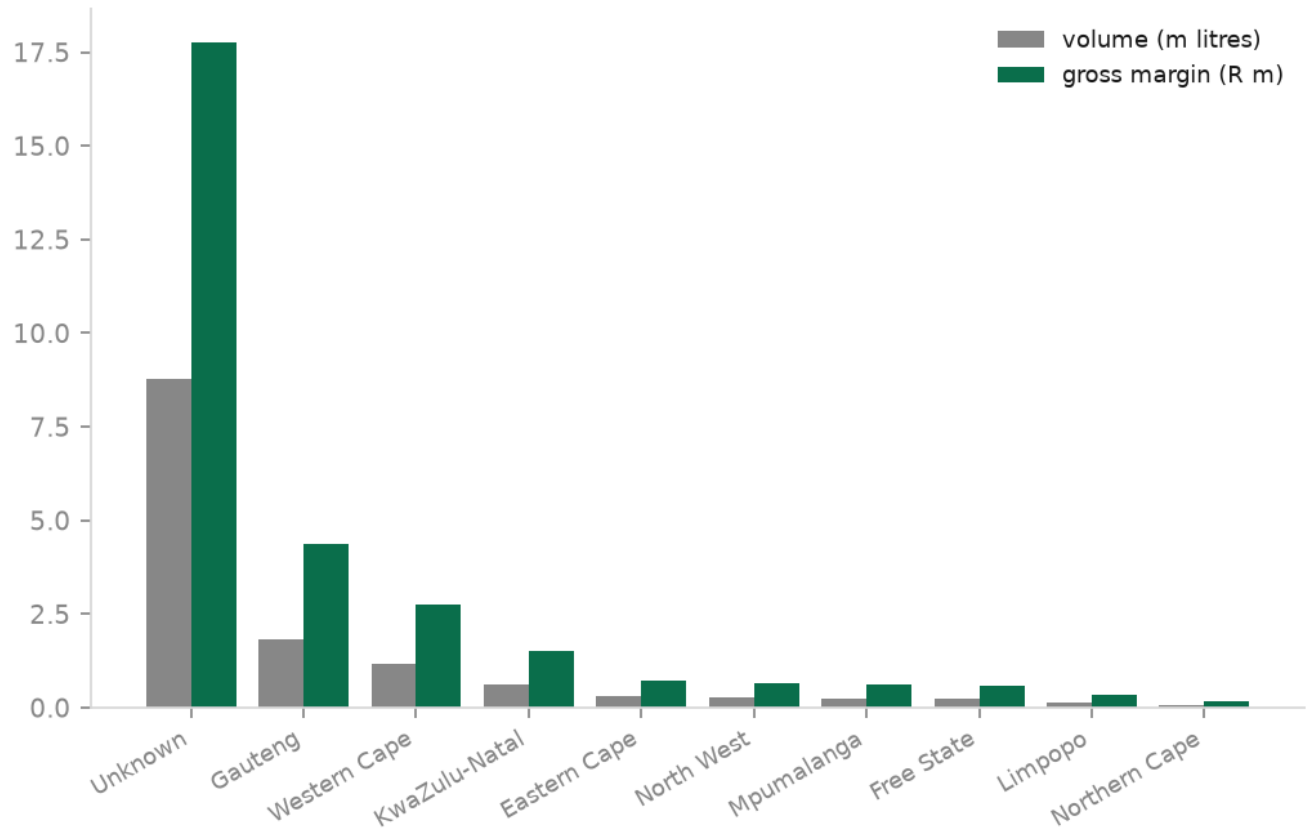
Plan by province

Province	Projects	Capex (R m)	Uplift (R m)
Gauteng	84	439.20	109.11
Western Cape	51	279.30	79.15
KwaZulu-Natal	34	188.00	44.93
Eastern Cape	16	87.30	21.48
Free State	16	85.60	19.52
North West	15	85.00	21.03
Mpumalanga	15	79.30	16.77
Northern Cape	7	42.20	11.74
Limpopo	9	38.20	8.19



The same network read six ways, on a fixed basemap and extent so the panels are directly comparable. A single map answers one question; a network planner asks six.

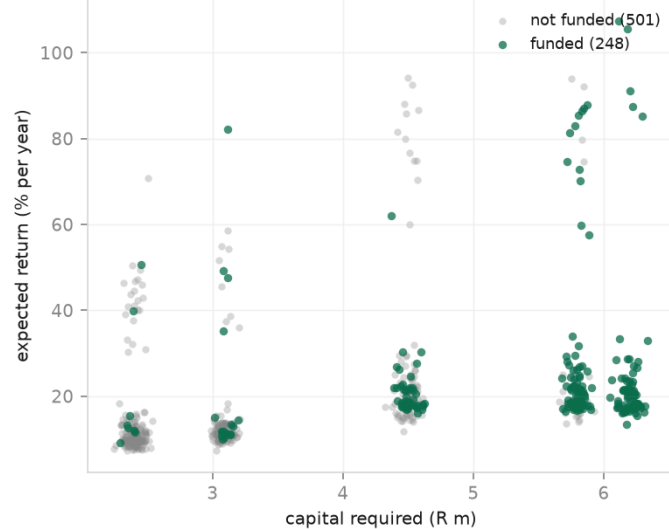
Volume and margin by province



Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

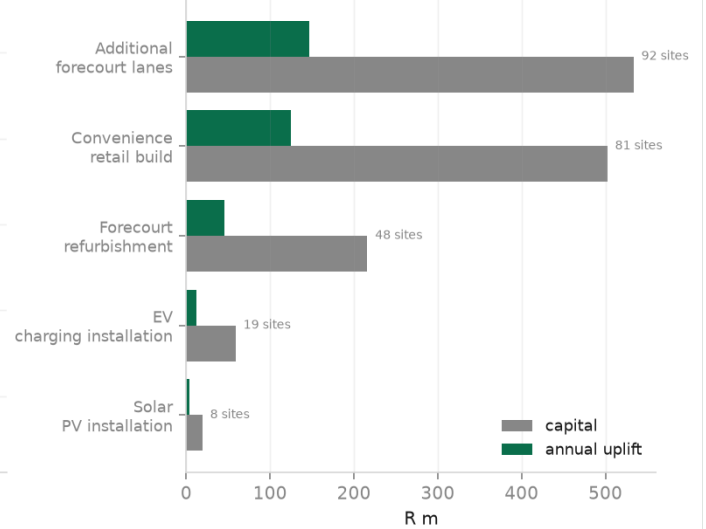
Volume and margin by province.

Candidate projects: return against capital

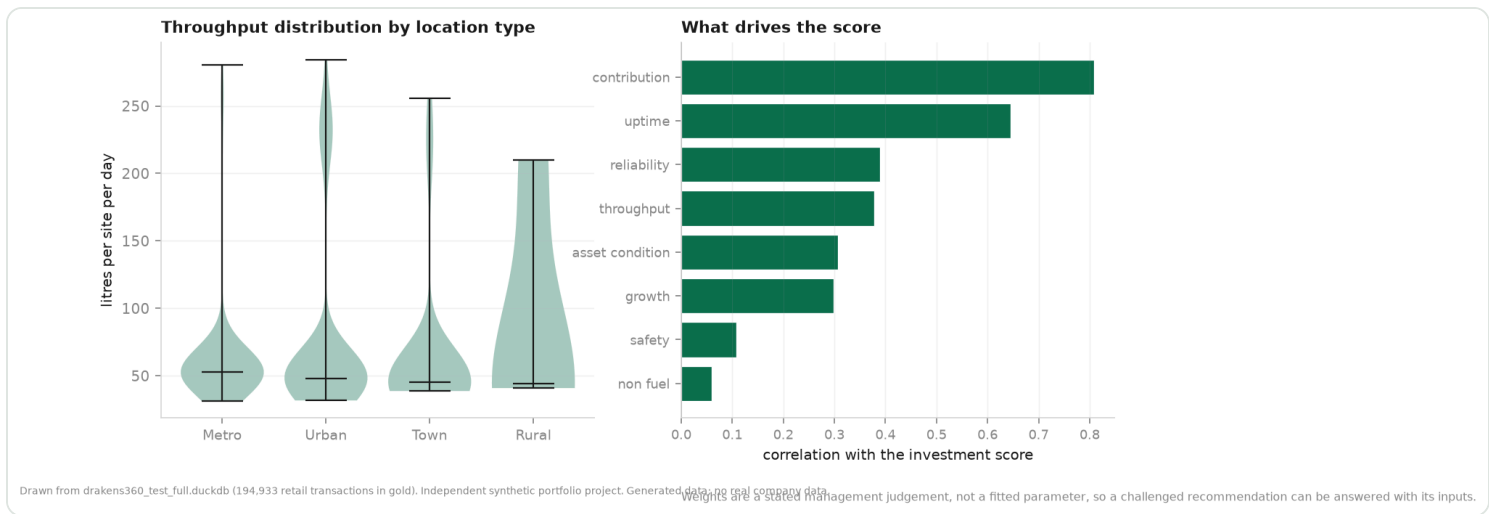


Drawn from drakens360_test_full.duckdb (194,933 retail transactions in gold). Independent synthetic portfolio project. Generated data; no real company data.

Funded plan by intervention



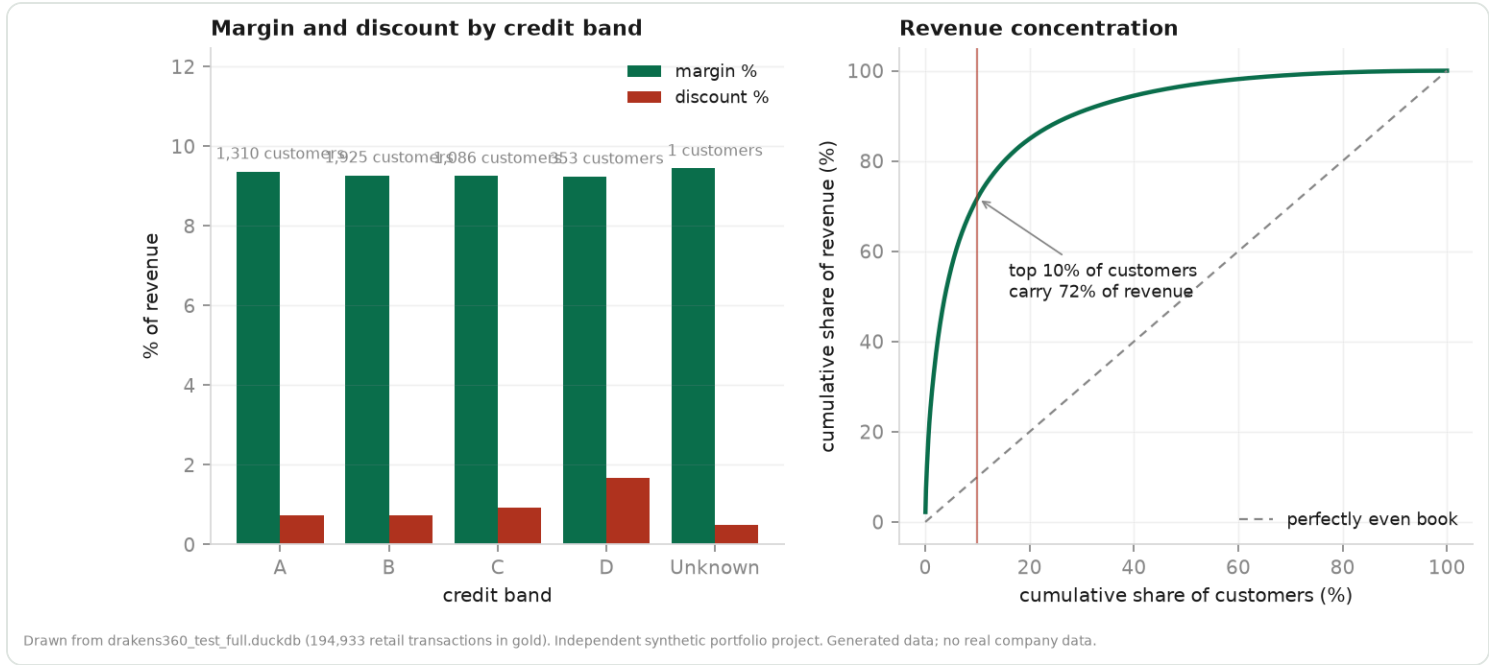
Every candidate project by return against capital, with the funded set highlighted. The funded set is not simply the top of the ROI ranking — provincial minimums and corridor protection pull specific projects in.



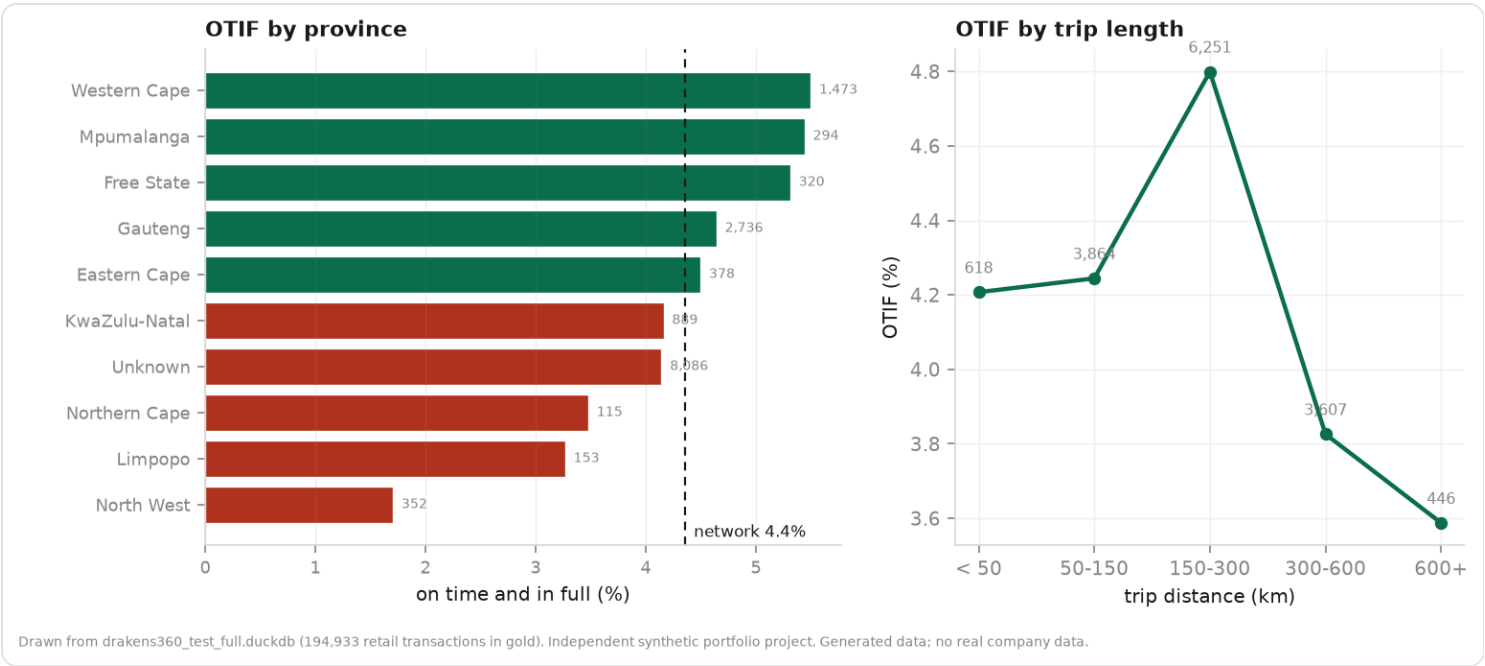
Left: throughput is skewed enough that a network average is a poor summary. Right: what actually drives the investment score, as the correlation between each component and the total.

Commercial and operations

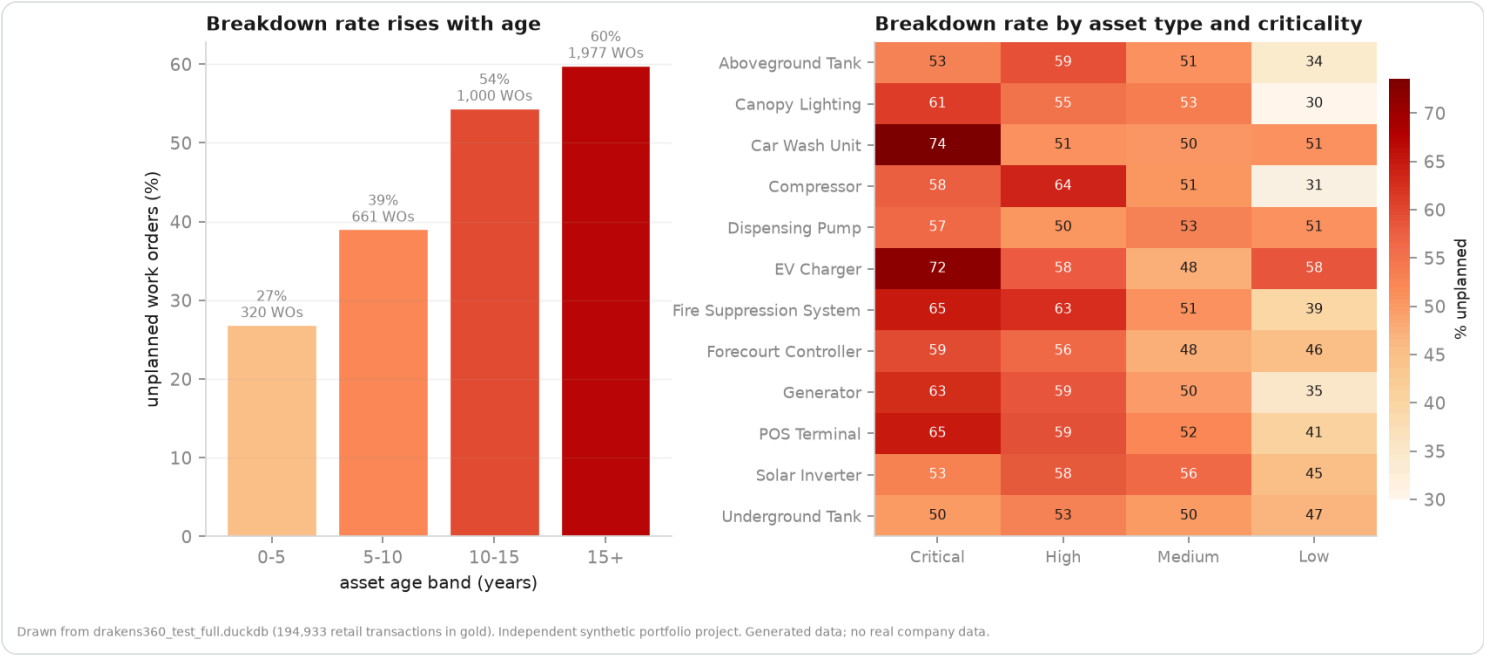
Four questions the business asks that the platform is built to answer, each computed once in gold rather than reassembled per report.



Left: whether the credit band means anything — worse-rated customers discount harder. Right: how concentrated the customer book is, which is a different risk from an even one.



OTIF broken out by province and by trip length. As a single network number it is the least useful form of it: broken out, it says whether lateness is a routing problem, a distance problem or one region’s problem.

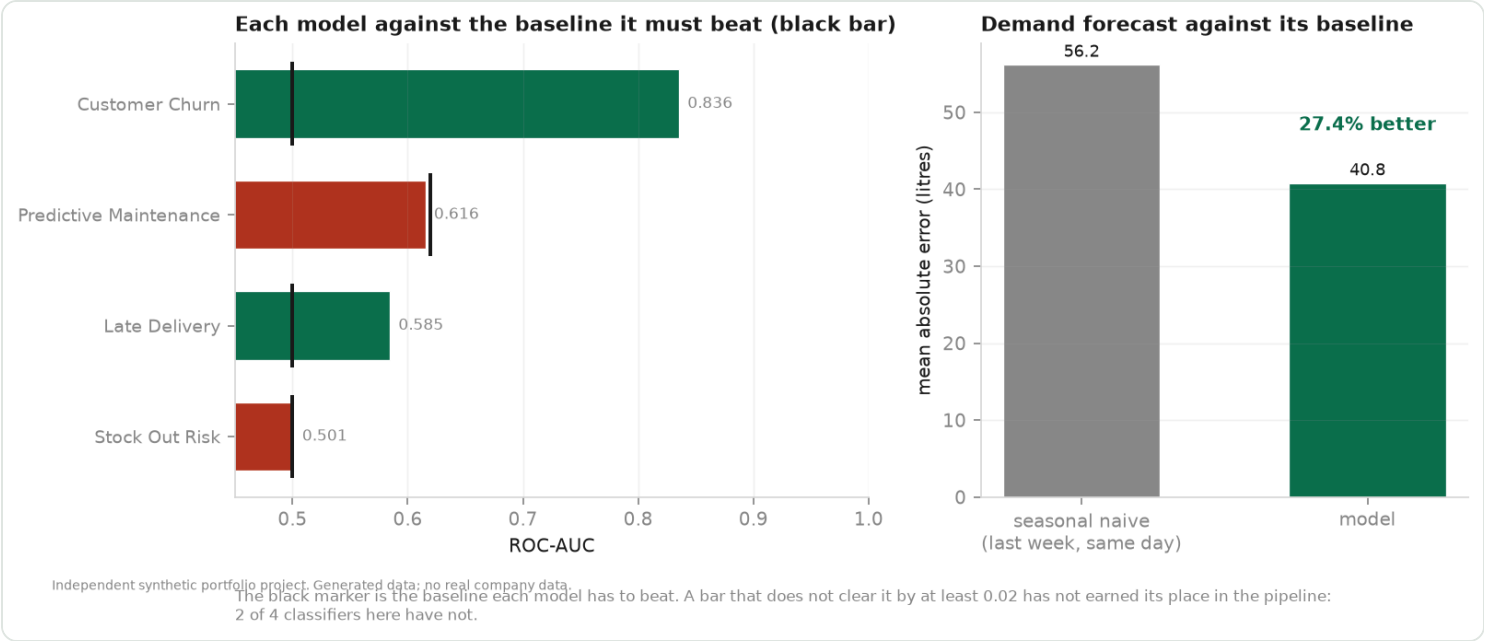


Breakdown rate against asset age and criticality. This is the signal the predictive-maintenance model is allowed to learn from, and the reason that experiment reports an age-only baseline.

Machine learning

Six experiments, each reported against a baseline it has to beat. Not all of them win, and all six are listed — a portfolio that shows only the models that worked is not showing the job.

Model	Headline result	Verdict
Customer Churn	ROC-AUC 0.836, 82% precision in the top decile	Usable — ranks well above chance
Demand Forecast	MAE 40.8 L against a seasonal-naive 56.2 L	Beats its baseline by 27.4%
Fraud Anomaly	1.0% of shifts flagged; cash share 30.9% against 23.9% for normal shifts	Flagged shifts differ from normal ones on both axes
Late Delivery	ROC-AUC 0.585, 1.72x lift in the top decile	Usable — the reviewed decile is enriched
Predictive Maintenance	ROC-AUC 0.616, 1.19x lift in the top decile, against a baseline of 0.620	Does NOT beat its baseline
Stock Out Risk	ROC-AUC 0.501	No usable signal



Each classifier against the baseline it has to beat, and the demand forecast against its seasonal-naive baseline. A bar that does not clear its black marker has not earned its place in the pipeline.

What the platform found in itself

Each of these was caught by the project’s own tests, observability or baselines, which is the point of having them.

A macro that silently corrupted every ratio

`safe_divide('a - b', 'c')` compiled to `a - (b / c)` because the macro did not parenthesise its arguments. Not an error — a plausible-looking wrong number. Caught by a range test asserting that `dim_product.base_margin_pct` falls between -5 and 100.

A general ledger that did not balance

Journal lines were posted independently, so debits exceeded credits by 46%. Caught by `obs_reconciliation_controls`, which evaluates double-entry as data rather than only as a test. Fixed by emitting journals as balanced pairs.

A partitioning choice that looked obvious and was wrong

The retail fact was partitioned on `date_key`, because every query filters on a date range. The maintenance job measured the result: 974 files averaging 0.2 MB, which OPTIMIZE can never merge because it does not cross partition boundaries. Under liquid clustering on `(date_key, site_id)` the same 5 million rows occupy 2 files averaging 94 MB — a 487-fold reduction in file count for identical data.

A leak that scored 0.998

The stock-out model was scored against the current inventory snapshot. Its target is `days_of_cover` below a threshold; `days_of_cover` is stock over usage; and both components were features. Excluding `days_of_cover`, which the code did with a comment explaining why, excluded nothing — the model divided one component by the other and reproduced the rule. Re-pointed at the next snapshot it scores 0.510.

A model with no baseline, losing to one variable

Predictive maintenance reported a 1.35x lift and passed review. It had no baseline. Ranking the queue by asset age alone scores ROC-AUC 0.620 against the model's 0.598, and the reported lift moves between 1.03 and 1.35 across capacity settings with no trend. Every experiment now computes and logs the baseline it has to beat.

Null foreign keys where an unknown member belonged

The landing zone contains referential drift by design — a site code in a fact that the dimension never received. Gold left-joined those facts and kept the null key, which drops the row from every inner join and from any total grouped by province: 5,070 retail transactions with real litres and real margin were invisible to regional reporting. They now resolve to a -1 unknown member, so nothing disappears from a total and the drift is countable.

A rule inferred from a column name, four times over

The pattern recurred often enough to be worth naming. A validity rule written against what a column is *called* rather than what it *means* destroys real data, and does it silently while every model builds and every test passes. ``signed_amount_zar >= 0`` quarantined every credit leg of the general ledger. A 2023-2027 timestamp window nulled 89% of site opening dates. An 8-45 rand fuel price band rejected 36% of the shop feed, where the same column name carries the price of a pie. Requiring a site on every row quarantined 58% of digital events, which are app sessions and were never at a station. Bands are now keyed by table and column, attribute dates get a window wide enough for a business history, and optional foreign keys are measured rather than assumed.

A tab that split a dimension in two

The maintenance heatmap showed thirteen asset types where the dimension has twelve. `clean_text` called `trim()`, which in both DuckDB and Spark removes spaces and nothing else, so a tab-padded value reached gold as a separate category from the same value without one. Both real, both counted separately, neither obviously wrong in a total.

Observability that measured 8% of what it claimed

`obs_cleansing_summary` named fourteen tables against a landing zone of 167, and the fourteen were whichever tables someone had thought to add — the least useful sample available. It is now generated by the same pass that generates the cleansing models: 128 monitored tables, 2,049,277 raw rows reconciling exactly to cleansed plus quarantined plus deduplicated, and zero tables failing reconciliation.

A validity rule that deleted half the general ledger

The contract applied a blanket rule to every amount column: anything named `*_zar` must be non-negative. `signed_amount_zar` is negative on the credit side by design, so the rule quarantined 41,886 of the ledger's 84,000 lines — every credit leg of every journal. What survived was a ledger of debits with almost no credits, which reported a 99.98% imbalance that looked like a generator bug and was a contract bug. A validity rule inferred from a column name is a guess, and a wrong guess deletes data silently instead of failing loudly. Signed measures are now matched by naming convention and excluded, so a new `net_` or `_variance` column is covered the day it is added.

Double-entry integrity is a property of the journal, not the line

Cleansing assesses rows one at a time, so when one leg of a balanced pair failed its contract the other leg survived alone. Each orphaned leg sat in the ledger as an unmatched debit or credit, and the ledger could not balance however correct the generator was — the imbalance was created by the cleansing step itself. The mart now publishes only journals whose lines still net to zero, and the orphans stay in quarantine beside the leg that was rejected. Debits and credits now agree exactly, to the cent.

A sentinel that passed every non-negativity check

The generator uses 0 as one of its numeric sentinels, and 0 satisfies every 'amount ≥ 0 ' rule. Stripping zero the way -999 is stripped is not an option, because zero litres and zero margin are legitimate readings. The contract now checks amounts against their own row instead: a transaction of 82 litres at R22.34 reporting R0.00 of sales is internally inconsistent whatever caused it.

Limitations

Stated plainly, because a report that claims completeness is misleading.

- The Databricks results are at portfolio scale (5,000,000 retail rows through bronze, silver and gold). The DuckDB warehouse figures in this report are at test scale, and each figure is stamped with the warehouse it came from. The two should not be read as one run.
- Predictive maintenance does not beat an age-only baseline, and stock-out risk has no signal at the current snapshot cadence. Both limits are in the generated data rather than in the estimators, and both fixes are changes to the generator: condition monitoring for the first, denser inventory readings for the second.
- The investment plan's rand figures rest on one stated benchmark — what a mid-sized site earns in fuel gross margin in a year — because the transaction fact is a sample and capital costs are not. Rank order and every ratio between sites are unaffected; only the level is rebased.
- Row filters are correct but their query cost under concurrent executive load has not been measured.
- There is no data subject access or erasure workflow. The masking and pseudonym design supports one; the process itself is not built.

Drakens Energy is a fictional company invented for this project. It does not exist. Every site, customer, employee, supplier, asset, coordinate, price, volume, margin and incident in this report is randomly generated. Site coordinates are South African town centroids plus random jitter and do not represent any real service station.